

Exercise – The Stympthalian Birds

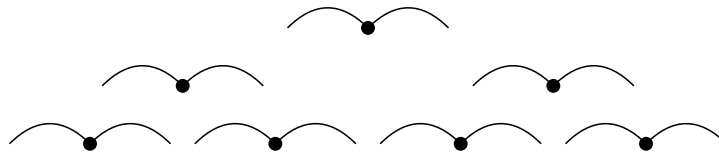
“These fly against those who come to hunt them, wounding and killing them with their beaks.”

Description of Greece, by Pausanias

The sixth labor of Hercules is to defeat the *Stympthalian Birds*. These man-eating birds brought up by Ares¹ have been ravaging the countryside of Arcadia, destroying crops, fruit trees, and harassing villagers.

Hercules quickly realized that he could not step into the swamps where the birds nest, as the ground would not support his weight. He cunningly used a rattle given to him by Athena² to frighten the birds and get them into the air. No sooner had he done that, the birds organized into a peculiar attack formation, ready to strike! There are $n = 2^i - 1$, for some $i \in \mathbb{N}$, birds $b_0, \dots, b_{n-1}$ in the formation, arranged as follows:

1. birds b_i , for $i \in \{\frac{n-1}{2}, \dots, n-1\}$, form the first row in this order (from Hercules' point of view);
2. bird b_i , for $i \in \{\frac{n-3}{2}, \dots, 0\}$, flies behind and between birds b_{2i+1} and b_{2i+2} .



The birds, being magical creatures raised by the god of war himself, have a built-in killer instinct, which can trigger in two ways: (1) if Hercules tries to take down the bird b_i before having taken down *both* birds b_{2i+1} and b_{2i+2} in front of it³; or (2) after a certain number t_i of seconds have passed. Should either (or both) of these conditions apply to the bird b_i , it goes into a *killing plunge* towards whomever is in their sight.

Hercules, armed with a bow and arrows whose tips have been dipped into poisonous blood of the slain Hydra, can take down *exactly one bird per second*. He fires his arrows in succession: the initial arrow takes a bird down at time 0 and every following second another arrow takes another bird down. Can he take down all the birds before any of them go into a killing plunge? Note that the bird b_i has to be taken down by an arrow *strictly before* time t_i .

Input The first line of the input contains the number $t \leq 50$ of test cases. Each of the t test cases is described as follows.

- It starts with a line that contains one integer n , denoting the number of birds ($n = 2^i - 1$, for $i \in \{1, \dots, 16\}$).

¹In Greek mythology, Ares is the god of war and courage, one of the twelve Olympians, and the son of Zeus and Hera.

²In Greek mythology, Athena is the goddess of wisdom, handicraft, and warfare, one of the twelve Olympians, and the daughter of Zeus.

³This cannot happen to the birds in the first row because they do not have any birds in front of them.

- The following line defines the individual killing plunge timers of the birds. The line contains n integers $t_0 \dots t_{n-1}$, separated by a space, and such that $1 \leq t_i \leq 2^{30}$, for $i \in \{0, \dots, n-1\}$.

Output For each test case output a single line with the string “yes” if Hercules can take down all the birds without any of them going into a killing plunge, and otherwise output the string “no”.

Points There are four groups of test sets, each worth 25 points. So, there are $4 \cdot 25 = 100$ points in total.

1. For the first group of test sets, you may assume that there are no more than seven birds ($n \leq 7$).
2. For the second group of test sets, you may assume that there are no more than $5 \cdot 10^3$ birds ($n \leq 5 \cdot 10^3$) and that the killing plunge times respect the “fly behind and between”-order (for all $i \in \{0, \dots, \frac{n-3}{2}\}$: $t_{2i+1}, t_{2i+2} < t_i$).
3. For the third group of test sets, you may assume that there are no more than $5 \cdot 10^3$ birds ($n \leq 5 \cdot 10^3$).
4. For the fourth group of test sets, there are no additional assumptions.

Corresponding sample test sets are contained in `testi.in/out`, for $i \in \{1, 2, 3, 4\}$.

Sample Input

```
2
7
7 6 5 1 2 8 9
3
3 1 1
```

Sample Output

```
yes
no
```